

Xudong Zhu

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I am a Ph.D. student in Computer Science at The Ohio State University, advised by Prof. Zhihui Zhu. My research focuses on representation learning and training dynamics in large language models, using these perspectives to advance mechanistic interpretability. In particular, I study linear phenomena in model representations: how a concept propagates during generation and whether it stays linearly represented along the way, and how training methods such as SFT and RLVR shape these structures. Before OSU, I received my B.S. in Computer Science from the University of Electronic Science and Technology of China (UESTC), where I worked with Prof. Zhao Kang on document-level relation extraction and Prof. Dong Hao on graph learning.

RESEARCH INTERESTS

Representation Learning · Interpretability · Representation & Behavior Steering · Training Dynamics · Agent Behavior Control

EDUCATION

The Ohio State University Ph.D. in Computer Science and Engineering, advised by Prof. Zhihui Zhu	2024 – Present Columbus, OH
University of Electronic Science and Technology of China B.S. in Computer Science · GPA 3.98 / 4.00	2020 – 2024 Chengdu, China

PUBLICATIONS

AbsTopK: Rethinking Sparse Autoencoders for Bidirectional Features

Xudong Zhu, Mohammad Mahdi Khalili, Zhihui Zhu

[ICLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)

From Emergence to Control: Probing and Modulating Self-Reflection in Language Models

Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, Zhihui Zhu

[Transactions on Machine Learning Research \(TMLR\)](#) · [Paper](#) · [Code](#) · [Project Page](#)

FCDS: Fusing Constituency and Dependency Syntax into Document-Level Relation Extraction

Xudong Zhu, Zhao Kang, Bei Hui

[LREC-COLING 2024](#) · [Paper](#) · [Code](#) · [Project Page](#)

Alleviating Subgraph-Induced Oversmoothing in Link Prediction via Coarse Graining

Xudong Zhu, Dong Hao, Ziqin Gao, Liming Pan

[Neurocomputing, 2025](#) · [Paper](#) · [Code](#) · [Project Page](#)

HONORS & RESEARCH SUPPORT

MMLS 2026 Travel Award , Midwest Machine Learning Symposium	2026
NAIRR Pilot Project NAIRR260106 (contributor), National AI Research Resource Pilot Geometry and Training Dynamics of Representations in Large Language Models	2026

OPEN SOURCE

LLMUsage · Local-first LLM subscription usage monitor for Hyprland (OpenAI, GLM, Grok, Claude)	TypeScript
dotfiles · fish, Neovim, and OpenCode configuration managed with GNU Stow	Lua